

HA LE

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TECHNICAL SKILLS

- **Languages:** TypeScript, JavaScript, Python, SQL, C/C++, Bash
- **Frontend & Mobile:** React, React Native (Expo), Next.js, Redux, Expo Router
- **Backend:** Node.js, Koa, Express, REST APIs, Knex/Objection ORM, Redis, MeiliSearch, Supabase
- **Data & Cloud:** PostgreSQL, BigQuery, AWS, GCP, Docker, GitHub Actions, EAS
- **Auth & Integrations:** OAuth2, SSO (ClassLink), Stripe, RevenueCat, Mux, PostHog, Klaviyo

PROFESSIONAL EXPERIENCE

Software Developer | *Vooks, Inc. (children's streaming platform)* May 2025 – Present

- Contributed **1,100+ commits** and **265 PRs** across 6 production repositories spanning React Native (Expo), Node.js/Koa, Next.js, and an admin panel, resolving **100+ tracked issues**
- Built **school SSO via ClassLink OAuth**, including tenant onboarding, domain verification, and the backend token exchange, giving districts single sign-on access to the platform
- Architected a cross-platform **recommendation and auto-play engine** on iOS, Android, and web, and shipped an **audio library** end to end with playback, listen-time tracking, Mux analytics, and MeiliSearch ranking
- Own the **analytics platform**: BigQuery pipelines, Looker Studio dashboards, client and server event tracking, and a nightly data-integrity job that detects gaps across five reporting datasets and alerts on regressions
- Built **subscription and payments** flows across Stripe and RevenueCat, and resolved production performance issues including **N+1 query elimination**, Redis caching, and PostgreSQL index tuning

Software Engineer (Contract) | *Redwood EDA* Jun 2026 – Present

- Built **Conversion Atlas**, the web console for an LLM-driven code conversion pipeline: steppable run timeline, inline and side-by-side diffs, summary dashboard, and per-attempt model exchanges
- Designed a **per-step model router** that replaced a single-model agent loop with cheap-first API calls and automatic escalation on verification failure, cutting the cost of a full module conversion from **\$8.22 to \$0.60** at equal verification coverage
- Added an **LLM reviewer stage** that gates each step on intent while formal equivalence gates correctness, catching silent no-op submissions that formal checking passes by construction
- Benchmarked **10 models** across six providers on identical conversion tasks and published the cost, completion, and failure data; contributed **15+ merged pull requests** across two open-source repositories

PROJECTS

Project Ava: Autonomous Verification Platform | *Python, Supabase, Fly.io, Netlify* projectava.dev

- Full-stack platform where an autonomous agent generates hardware testbenches, self-corrects through iterative simulation, and streams results to a live web dashboard backed by Supabase
- Verified **20 designs** across 191 tests; built a mutation testing engine covering 546 mutants in 7 categories

AgentTrace: Workload Profiling Dashboard | *Python, ECharts* agenttrace.quest

- Interactive analytics dashboard with Gantt timelines, convergence curves, and per-iteration latency charts; measured **43.3% idle time** and **3.6x** call amplification in an agentic AI pipeline

Open-Source Hardware | *TL-Verilog, Python, Yosys* github.com/vanthienha199

- Optimized a RISC-V core to the **fewest cells, fewest flops, and highest clock frequency** of four competing implementations in the same toolchain; taped out a separate agent-generated design on an open silicon shuttle

EDUCATION

University of Central Florida Orlando, FL

Bachelor of Science in Computer Science, GPA 3.52 May 2026

Research assistant, Unary Lab: multi-vendor GPU profiling tooling in Python and C++

University of South Florida Tampa, FL

Bachelor of Arts in Economics December 2023